

Worksheet No. - 2

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Aim/Overview of the practical:

To implement and analyse SQL SELECT queries using filtering, sorting, grouping, and aggregation concepts in PostgreSQL for efficient data retrieval and analytical reporting.

Objective:

- To retrieve specific data using filtering conditions
- To sort query results using single and multiple attributes
- To perform aggregation using grouping techniques
- To apply conditions on aggregated data
- To understand real-world analytical queries commonly asked in placement interviews

Input/Apparatus Used:

- PostgreSQL
- pgAdmin

Procedure/Algorithm/Code :

```
CREATE TABLE Students (  
    student_id INT,  
    name VARCHAR(50),  
    city VARCHAR(50),  
    percentage DECIMAL(5,2)  
);  
  
INSERT INTO Students VALUES  
(1, 'Amit', 'Delhi', 96.5),
```

```
(2, 'Riya', 'Mumbai', 94.2),  
(3, 'Rahul', 'Delhi', 97.8),  
(4, 'Sneha', 'Mumbai', 98.1),  
(5, 'Ankit', 'Chandigarh', 95.6),  
(6, 'Pooja', 'Delhi', 93.4),  
(7, 'Karan', 'Chandigarh', 96.2);
```

---Without Case Statement

```
SELECT CITY , COUNT(*) AS STUDET_COUNT FROM Students  
WHERE percentage> 95  
GROUP BY city;
```

-- WITH CASE STATEMENT

```
SELECT CITY, SUM(CASE WHEN percentage> 95 THEN 1  
ELSE 0 END) AS STUDENT_COUNTS FROM Students  
GROUP BY city
```

-- (II)

```
SELECT CITY, AVG(CASE WHEN PERCENTAGE>95 THEN PERCENTAGE ELSE  
NULL END) AS STUDENT_AVG FROM Students  
GROUP BY city  
ORDER BY STUDENT_AVG DESC;
```

Output:

	city character varying (50)	studet_count bigint
1	Delhi	2
2	Mumbai	1
3	Chandigarh	2

Without Case Statement

	city character varying (50)	student_counts bigint
1	Mumbai	1
2	Delhi	2
3	Chandigarh	2

With Case Statement - I

	city character varying (50)	student_avg numeric
1	Mumbai	98.1000000000000000
2	Delhi	97.1500000000000000
3	Chandigarh	95.9000000000000000

With Case Statement - II

Learning outcomes (What I have learnt):

1. I have understood how data can be filtered to retrieve only relevant records from a database.
2. I have learned how sorting improves the readability and usefulness of query results in reports.
3. I have gained the ability to group data effectively for analytical purposes.
4. I have clearly learned to differentiate between row-level conditions and group-level conditions.
5. I have developed confidence in writing analytical SQL queries used in real-world scenarios.
6. I have become better prepared to answer SQL-based placement and interview questions related to filtering, grouping, and aggregation.